



STUDY GUIDE

Internet Technologies

(Erasmus: Web programming)

Linas Būtėnas

Internet Technologies

Study Guide

Cycle: First

Study program: Internet Technologies

Course unit code: ITWEB

Awarding institution: Department of Computer Science II, Faculty of Mathematics and Informatics

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Preparation of the study guide was supported by the project „Increasing Internationality in Study Programs of the Department of Computer Science II“, project number VP1–2.2–ŠMM-07-K-02-070, funded by The European Social Fund Agency and the Government of Lithuania.

Studijų vadovo medžiagos rengimą rėmė projektas „Kompiuterijos katedros studijų programų tarptautiškumo didinimas“, projekto kodas VP1–2.2–ŠMM-07-K-02-070, finansuojamas iš Europos socialinio fondo ir Lietuvos valstybės biudžeto lėšų.

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1 Introduction

The aim of the course “Internet technologies” is to assist students in obtaining base knowledge and abilities in Internet technologies domain. At the end of this course students must be able to know and apply the essential principals of Internet technologies, be able to build simple web system, using HTML, CSS, scripting language (JavaScript), server side scripting language (PHP) and web framework (PHP framework – CodeIgniter, Laravel).

Teaching and learning methods

In this course during lectures we will use interactive lectures, discussions and case analysis methods. All lecture notes, course material, tests, deadlines, links, assignment submissions and interactive communication (forums, news) will be carried out using learning content management system of Vilnius University (<https://emokymai.vu.lt/>).

The outcome of the course

After finishing the course, students should have these abilities:

- Ability to define the problem for the application area and apply proper existing solutions.
- Knowledge of main principles of server-client communication on the World Wide Web.
- Knowledge of main principles of markup and style languages (HTML, XML, CSS), possibilities of using them and ability to apply it.
- Ability to write programs using server-side and client-side scripting languages
- Ability to analyze and understand program written in scripting language, its structure; ability to execute a program and change program code
- Ability to analyze web framework structure, its working principles and perform necessary framework code changes to accomplish given task
- Ability to apply information security mechanisms in web systems development

How to use this guide

You will find the course content divided in to several parts:

- **Content** - Describes the main themes of each topic.
- **To learn** - Shows the learning material, which is mandatory.
- **Optional** - This is optional learning material. Read, review or practice it if you already know the mandatory material, have some knowledge gaps in this theme or you feel you have to strengthen your knowledge. The knowledge of these themes is not tested during evaluation.
- **Tasks for practice sessions and self-assessment assignments** - Here the recommended plan for practice sessions is given. There are also given assignments for knowledge and abilities development and self-evaluation.
- **Guidelines for individual work**
 - If you are unsure, what you have to do during self-study time, you can follow the given guidelines in this section.
 - **(Project assignment)** – this part is dedicated for helping students to plan tasks for project assignment fulfillment.

2 Course content

2.1 Communication principles of Web server and client

- URI, URL and URN
- Web server communication with a client, request/response cycle
- The general process of data flow between server and client
- Visualization of HTML (interpretation and its “place”, visualization, etc.)
- Execution of various script languages (JavaScript, PHP, etc.)
- Apache, Database, PHP packages – WAMP, XAMPP, MAMP

To learn:

- Wikipedia. “Uniform resource identifier”,
http://en.wikipedia.org/wiki/Uniform_resource_identifier
- “What really happens when you navigate to a URL”, <http://igoro.com/archive/what-really-happens-when-you-navigate-to-a-url/>
- **Video:**
 - What happens when you type a URL into your browser?
<https://www.youtube.com/watch?v=AlkDbnbv7dk>
 - HTTP 1 Vs HTTP 2 Vs HTTP 3!
<https://www.youtube.com/watch?v=UMwQjFzTQXw>
 - Top 6 Most Popular API Architecture Styles
<https://www.youtube.com/watch?v=4vLxWqE94l4>

Tasks for practice sessions and self-assessment assignments

Prepare initial space for web development (using faculty infrastructure)

or

Prepare initial space for web development (using your own computer)

If you are **using your own computer**, you have to install web server (for instance: Apache), PHP and database. It is much easier to install everything using installation and management packages like WAMP, XAMPP, MAMP, etc. These packages include Apache, PHP, MySQL (as database), and configures web server for instant usage.

- After installation local web server will be accessible through these links: <http://localhost/> or <http://127.0.0.1/>
- Create a sample HTML file (*index.html*) with some content, place it in new folder inside main web server *root* folder (it can be named *www*, or *htdocs*, or similar – read the manual of the installed system) and access it through web browser. To access your folder, go to such address: <http://localhost/<yourNewFolderName>/>
- *Hint:* To keep an order in your *root* folder, it is recommended that for each separate assignment you should create a separate folder. It is convenient, because each assignment/project has *index.html* or *index.php* file, *style.css* file, and many other files, additional libraries, script files with the same standard names as other projects.

2.2 Principles of HTML markup language

- Principles of HTML and XHTML
- HTML5 features

To learn:

- HTML5 tutorial <http://w3schools.com/html/default.asp>

Tasks for practice sessions and self-assessment assignments

Tasks described in part 2.3 (together with CSS tasks).

2.3 Principles of CSS and CCS frameworks

- CSS syntax, selector types, CSS box model
- Principle of cascading styles
- Interpretators of styles in web browsers
- Differences of CSS1, CSS2, CSS3
- CSS frameworks

To learn:

- CSS tutorial <http://www.w3schools.com/css/default.asp>
- CSS Float tutorial:
 - <https://internetingishard.netlify.app/html-and-css/floats/>
 - <http://css.maxdesign.com.au/floattutorial/>

Tasks for practice sessions and self-assessment assignments

- create HTML and CSS files from scratch
- take [Zen Garden](#) HTML and CSS templates and modify them
- take one of the open source web designs (for instance: <http://www.oswd.org> or <http://www.opendesigns.org>) and modify it

The final goal is to have HTML and CSS files demonstrating:

- selection of HTML elements by tag names, classes and id's
- style combining feature, using several classes in one HTML object
- using multiple classes in HTML elements
- style cascading feature
- several elements using CSS float property
- using at least 2 properties from 5 CSS property groups (example: 2 properties from 1 CSS property group: border-color and border-style)

Optional:

- Read about several CSS frameworks and try out at least one.
- Good web sites about CSS frameworks:
 - <https://github.com/troxler/awesome-css-frameworks>
 - <https://www.geeksforgeeks.org/css-framework/>

Guidelines for individual work

- **Project assignment:**
 - Decide what would be the main idea of your web system. An example can be: a blog, a forum, e-shop, social communication system, web gallery, etc.
 - Create a sample of the first web page of web system in HTML and CSS. You will have to use it later.

Hints

- Do not forget that HTML comments are visible to user through “View HTML source” browser functionality. So, do not leave comments revealing code functionality or direct access to restricted resources (as links to administration pages or hidden functionality).

2.4 Programming using JavaScript and JavaScript libraries/frameworks

- History of JavaScript
- Basic principles of JavaScript
- JavaScript grammar and use cases, JavaScript dialects
- JavaScript and HTML DOM relation
- JavaScript virtual machine, debugging
- AJAX
- JavaScript libraries (jQuery)

To learn:

- JavaScript tutorial: <http://www.w3schools.com/js/default.asp>
- jQuery tutorial <http://www.w3schools.com/jquery/default.asp>

Tasks for practice sessions and self-assessment assignments

Create a HTML file with some JavaScript code:

- Create your own JavaScript function with parameters
- Use JavaScript Array object
- Call your function, where function parameter value can be set dynamically from web page
- Chose 3 HTML DOM objects and use at least 1 of their property/method
- Go through given JQuery tutorial and try out all given tasks and examples

Guidelines for individual work

You can try out advanced features of JavaScript or JQuery:

- Dynamically change HTML page style switching the CSS files using JavaScript
- Using JQuery implement Ajax functionality, demonstrating how the content of the page can be dynamically changed by new source received from the server
- **Project assignment:**
 - If you plan to use JavaScript in your project, think which parts of HTML page would be dynamically changed or animated
 - You can already build up your JavaScript (jQuery) code on the prepared HTML page. For instance, you can validate values in the input fields, make a folding menu,

use Ajax for dynamic changes in web page content, etc.

2.5 Programming web systems using scripting language (PHP)

- Basic principles of web programming
- Differences of scripting and compiled languages
- PHP features and syntax
- Most important differences from C or JAVA
 - dynamic variable types, multi-dimensional dynamic arrays, global and predefined variables, variable variables, variable casting, call-back functions, debugging, etc.
- Data passing client → server → PHP script
- Data storage, session, cookies

To learn:

- PHP manual (can be used as tutorial) <http://php.net/manual/en/index.php>
- PHP tutorial: <http://www.w3schools.com/php/default.asp>

Tasks for practice sessions and self-assessment assignments

- Create an *index.php* page and view it using the internet browser
- Create one class and 2 functions
- Use multi-dimensional array
 - Create user interface for handling array data: printing the elements, adding new elements, deleting elements
- Use one of cycle control structures (while, for, foreach)
- Define HTML form <form>, with POST or GET method. Send several values of input fields to other web page, where these values will be displayed
- Use session variable (\$_SESSION) to store values between different page requests
- Use cookie (setting at client side, reading values of stored cookie)
- Use database to store data. Create data adding, updating, deleting functionality

2.6 Principles of web frameworks, their usage and application

- Web framework basic principles and features
- Architectures of frameworks (MVC, module based)
- Main parts, functions of frameworks (configuration, libraries, multi-language, user handling/security, etc.)

To learn (for beginners):

- CodeIgniter <http://www.codeigniter.com/>
 - tutorial https://www.codeigniter.com/user_guide/intro/index.html

Tasks for practice sessions and self-assessment assignments

- Download web framework and “make it work”
- Create a simple web page having logical parts (header, menu, body and footer) and implement them as different views
- Create a user functionality:
 - user registration form
 - user login form
 - user data view page

Guidelines for individual work

Project assignment:

- Prepare web framework for project development
 - Adjust project configuration in *config* folder. Read about CodeIgniter Routing and URL building. There is also section 5.1 in this study guide about URL building.
- Use already prepared HTML and CSS files for main page creation
 - Split your page content logically to header, contents and footer sections.
- Plan what functions and features your project will have and start to implement them.
- All required features for project implementation and defense are given in section 3.1

2.7 XML language, main usage principles and application

- XML standard history, purpose and related standards
- Main principles and syntax of XML
- Well-formed XML and validation
- Overview of DTD, XPath, XSD, XSL

To learn:

- XML <http://www.w3schools.com/xml/default.asp>

Tasks for practice sessions and self-assessment assignments

- Create XML structure of 6 elements (types) and describe it with XSD (XML Schema).
- Elements should be complex elements:
 - 1 element has other elements inside;
 - 1 element has to be empty;
 - 2 elements has to be XSD built-in type (xs:integer, xs:date, ...) with restrictions/facets;
 - 1 element has to be xs:pattern type;
 - 1 element has to have an attribute.
- Create at least 2 your own data types, in such a way that one of them inherits the other. One of the XML file elements has to use one of these types.
- The same XML structure define using DTD
- Create XML file, having at least 2 records. Each defined element in XSD should be used at least once.
 - XML validator should confirm that XML file is valid according to XSD and DTD
 - Try to make different changes in XML file and test what feedback/view gives internet

browser and XML file validator

- Create XSL file for XML file:
 - Use <xsl:for-each>, <xsl:value-of>, <xsl:if> (or <xsl:choose>) elements;
 - Use at least one XSL function;
 - Use <xsl:value-of> to get elements value and attribute value

Example:

```
<?xml version="1.0" ?>
<notes>
  <note>
    <to> President </to>
    <from> Vice-president </from>
    <body> Lets meet at 10:00 in my office </body>
  </note>
  <note>
    <to> Ela </to>
    <from> Tom </from>
    <body> Hello! </body>
  </note>
</notes>
```

3 Requirements for evaluation

3.1 Overall evaluation strategy and requirements

The final grade consists of several parts:

Activity	Grade	Detailed description of requirements
Project (and its defence)	6	Section 3.2
Work during practice class	0,5	Section 3.3
Exam	3,5	Section 3.4
Total:	10	

3.2 Project assignment and project defence

Maximum grade: 6

For project defence students have to create a web site using framework.

- Recommended framework: CodeIgniter (for PHP)

Table 1. Requirements with grades.

Requirements	Grade
General	pass/fail
Configuration	0,5
HTML, Templates	0,5
Design, CSS	0,5
Object management	1
Validator	0,5
User login and rights	1
Database and MVC Model	1
Logging & Good practice	0,5
System deployment	0,5
Total:	6

Requirements for web site creation:

- **General**
 - Website contents are displayed in one language (Lithuanian / English)
 - All web pages must follow the same design pattern
 - You have to be able to make changes in program code during project defence
 - Website is created using web framework
 - System is deployed on MIF Cloud infrastructure

- Code must be in MIF Git (<https://git.mif.vu.lt/>)
- **Configuration**
 - Use framework configuration possibilities (base_url (you must use it), routes, autoload, database, seeds for password generation, ...)
 - Use at least 5 configuration features/parameters
- **HTML, templates**
 - HTML structure must be valid
 - HTML code should be logically divided and implemented in different “views” (as header, content, footer, etc.)
- **Design, CSS**
 - Website should look at least minimally in order (imagine that you will have to show your solution to other IT students in other university). You can use free design templates from the internet, but you have to make your own CSS and HTML adjustments.
 - At least 10 CSS properties (property groups) should be used
 - You must use CCS framework
- **Object Management**
 - Website functionality for managing the objects. Each object must have standard management CRUD functions (add, view, edit, delete, view list)
 - At least 2 objects must be created
 - At least 8 properties/fields must exist per all objects (foreign key does not count)
 - User object or related objects (rights and groups) does not count as an object.
- **Validator**
 - All input fields has validators
 - Server side validator checks if input value satisfies conditions (before writing it to database). At least half of conditions must be not check only for empty field data, should check for particular data type/pattern.
 - Each error message is displayed in such way that it is clear which property has an error
 - Backend validators errors must be visible in front end (at least in 3 different cases)
 - At least 8 backend validators must be created
- **User Management**
 - User login functionality is implemented
 - At least 2 user types with different rights are implemented (for example: admin and ordinary user)
 - Both user types can login in to the website
 - The rights of the user types must differ
 - Logged in user must be able to complete 2 functions, inaccessible to not logged in users
 - Admin must have 2 functions, inaccessible to logged and not logged in users
 - The password must be stored encrypted in database
- **Database and ORM Model**
 - Use relational database to store data
 - Create model in framework for each object. Each model must have functions for object data retrieval and manipulation (select, insert, update, delete functionality)
 - The primary key must exist in every table
 - One object or user table must contain 100 records, one other object must contain 10000 records
- **Logging & Good practice**
 - Logging
 - Errors (and actions) must be logged in to the database or files
 - Each log record is stored as separate record and should clearly declare which class, method, algorithm part created this log item and the message should be

- very clear
 - At least 5 different actions/errors must be logged this way
 - **Good practice**
 - After form reload then the error is displayed – leave fields with the data, do not clear them
 - Pagination or lazy loading of item lists, then there are more than 20-25 items
- **System deployment**
 - Project is deployed on MIF Cloud infrastructure using Docker or Podman

What is tested – Defended project fulfilment according to given requirements, ability to explain the code of program, and the principles of operation of created project, the principles and workflow of used web framework.

Each student defending a project must be ready to answer these questions during defence:

- Describe how framework works from the first call to main Class-method till the last line of output is generated and sent to client;
- Describe how authentication/authorisation procedure works, where the data about the logged-in user is stored and how this data/object can be accessed and changed;
- Describe how objects are mapped to Classes/Models if ORM (Object Relational Mapping) is used, or framework itself creates relations to database, or creates/generates framework part or database part of object storing/handling; Describe how each object field value is getting/setting its value from/to database;

Examples of possible project ideas

You can create a blog, a forum, an e-shop, a small web system for community communication and decision making, etc. Any system having functionality described in requirements will be suitable. For more ideas you can browse in online freelance web job markets and look what people in the real business world is looking for.

Project defence deadlines

Defense date	Maximum grade
14-16th weeks of semester ¹	Project defense grade * 100 % (max. 6)
until the end of exam	0
During re-examination	0

¹ – each student must register in project defense schedule.

3.3 Work during practice class

Maximum grade: 0,5

An individual work during practice class is estimated and evaluated. Each student must show up in at least several practice class sessions and demonstrate that he/she is working by himself/herself on the project. The grade is given if a teacher is confident that student shows progress during practice session, student is able to solve tasks individually, is able to learn and apply the knowledge in practice.

What is tested – Ability to work individually, analyze problems/situations, create solution (program code), relate it to the knowledge and theme of the course.

3.4 Exam

Maximum grade: 3,5

Students have to complete a test, consisting of 5-25 closed or open type questions, evaluated 0,1-0,5 grade each. The value of the question answer depends on the abilities level that are tested. The content of the exam is formed from the whole course content.

What is tested – ability to demonstrate knowledge and application of course material.

Examples:

- *Question:* CSS is clipping of the words Cascading Style Sheets. Give at least one example of “cascading” effect of CSS.
- Give several examples of semantic and non-semantic HTML 5 elements. Explain why you think they belong to such type.
- What are the essential advantages and disadvantages of using CodeIgniter framework for web project creation?
- You have a XML code:

```
<?xml version="1.0" ?>
<notes>
  <note>
    <to> Teacher </to>
    <from> Student </from>
    <body> My assignment </body>
  </note>
  <note>
    <to> Ela </to>
    <from> Tom </from>
    <body> Hello!
      <subbody> lala </subbody>
    </body>
  </note>
</notes>
```

Choose a correct answer:

- This XML is well formed
 - This XML is valid
 - This XML is well formed and valid
 - This code is not a valid XML code
-
- Does XML allow having text and elements inside other elements? (like in element <body>)?